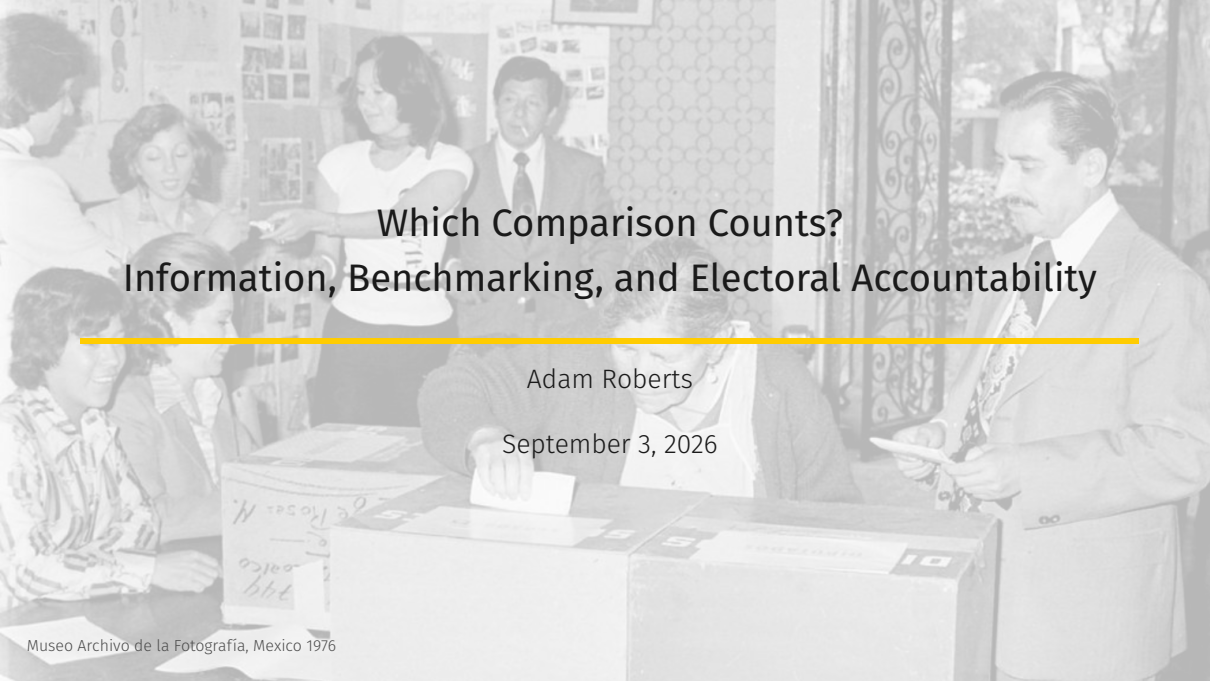




Which Comparison Counts? Information, Benchmarking, and Electoral Accountability

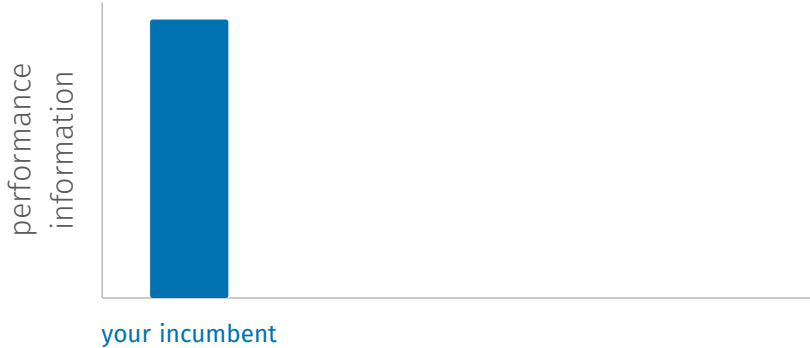
Adam Roberts

September 3, 2026

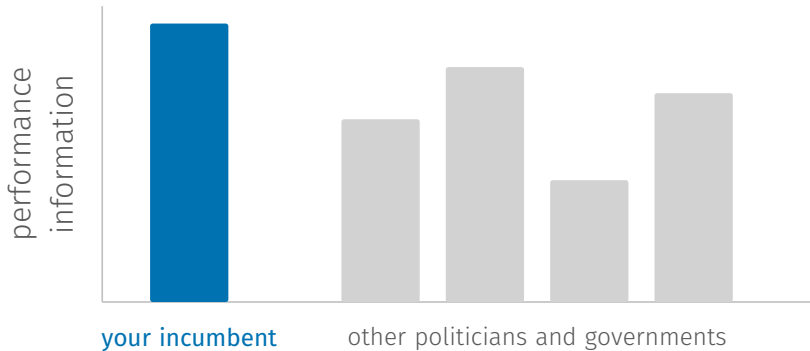
Electoral accountability in developing democracies

- Voting based on evaluations of politicians or other considerations?
- Unbiased information can be hard to obtain
- Experiments provide information, generally benchmarked for ease of interpretation
- How does benchmarking change how information is evaluated by voters?

Benchmarking information



Benchmarking information



Benchmarks bundle many design choices

1. Size of comparison group (whole country, state, or selected peers)
2. Nature of comparison group (co-partisan vs out-partisan vs non-partisan)
3. Presentation of information (text, table, graph)
4. Content of information (corruption vs. policy platforms)
5. Complexity of information (single vs. multiple metrics)

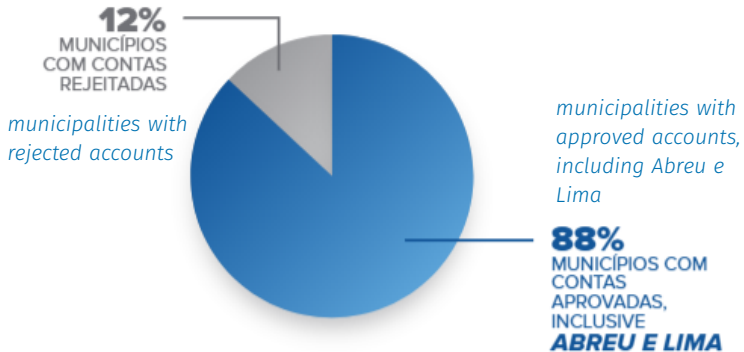


Figure 1: Informational leaflet from Boas, Hidalgo, and Melo (2019)

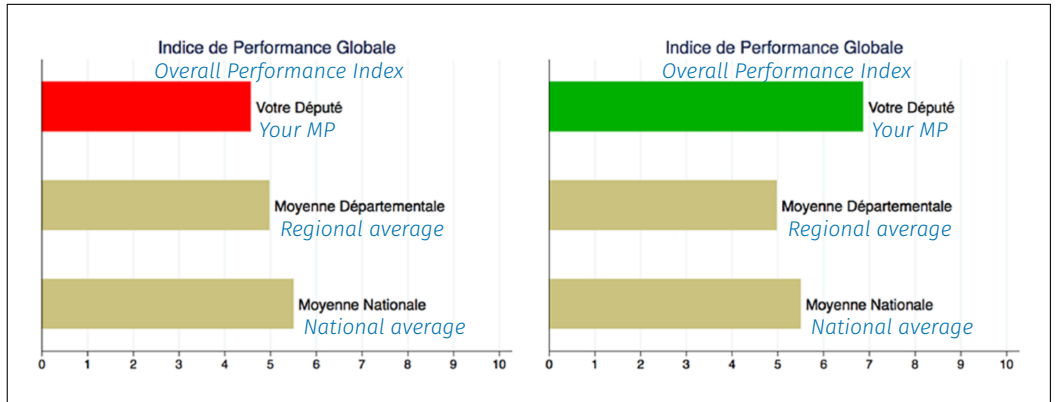


Figure 2: Informational leaflet from Adida et al. (2020)

EN 2013, EL **PARTIDO** QUE GOBIERNA **LA PAZ** RECIBIÓ **32.7 MILLONES** DE PESOS DEL FISM Y GASTÓ **0%** EN COSAS QUE **NO DEBE**

¡COMPAREMOS CON LOS GASTOS DE **OTROS PARTIDOS!**

MUNICIPIOS DE TU ESTADO GOBERNADOS POR **OTROS PARTIDOS** GASTARON EN PROMEDIO **9%** EN COSAS QUE **NO DEBEN**

Information from Arias et al. (2019)

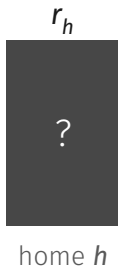
In 2013, the party governing La Paz received 32.7 million pesos from FISM and spent 0% on things it should not.

Let's compare it with the spending of other parties!

Municipalities in your state governed by other parties spent on average 9% on things they should not.

Observing crime rates

Observing the crime rate (r) conflates multiple underlying causes



Observing crime rates (cont.)

Observing the crime rate (r) conflates multiple underlying causes

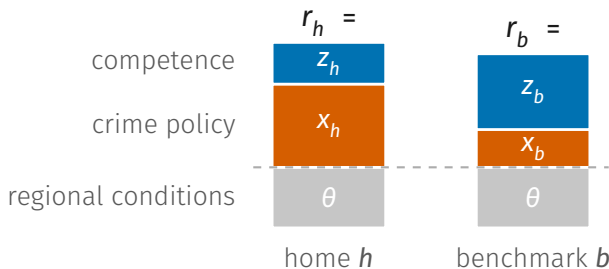
$$r_h =$$

competence	z_h
crime policy	x_h
regional conditions	θ

home h

Observing crime rates with a benchmark

Comparison to a benchmark differences out common factors



Many aspects of benchmarking

1. Size of comparison group (whole country, state, or selected peers)
2. Nature of comparison group (co-partisan vs out-partisan vs non-partisan)
3. Presentation of information (text, table, graph)
4. Content of information (e.g., corruption vs. performance)
5. Complexity of information (e.g., single vs. multiple metrics)

Comparing partisan benchmarks

$$d = r_h - r_b = (z_h - z_b) + (x_h - x_b) + (\theta - \theta)$$

Plain info

(no benchmark)

Non-partisan

z	z_h
x	x_h
θ	θ

nothing cancels

Δz
Δx

region cancels

Comparing partisan benchmarks

$$d = r_h - r_b = (z_h - z_b) + (x_h - x_b) + (\theta - \theta)$$

Plain info

(no benchmark)

Non-partisan

Out-coalition



nothing cancels

region cancels

policy cancels
depending on
voter priors

Comparing partisan benchmarks

$$d = r_h - r_b = (z_h - z_b) + (x_h - x_b) + (\theta - \theta)$$

Plain info

(no benchmark)

Non-partisan

Out-coalition

Same-coalition



nothing cancels

region cancels

policy cancels
depending on
voter priors

policy cancels,
competence
revealed

Empirical expectations for benchmarks

Voter beliefs	Informativeness of benchmark		
	Non-partisan	Opposition	Same-party
Parties alike	High	High	High
Differ, competence focus	Low	Prior-dependent	High
Differ, policy focus	None	High	None

Table 1: How informative each benchmark type is for the voter, under alternative voter beliefs about how parties differ.

Comparing benchmarks: survey in Mexico

Two components in one survey:

1. **Descriptive:** respondents select municipalities they consider relevant comparisons
2. **Experiment:** experimentally vary how crime information is benchmarked

Context:

Survey deployed in subset of states holding municipal elections in 2027

Provide real performance information, electoral parties/coalitions

Four treatment arms vary benchmarks shown

	Arm	What the respondent sees	N
	Control	Placebo text (rainfall statistics; no crime data)	394
T1	Plain info	Municipality robbery statistics	186
T2	Non-partisan	Plain info benchmarked by similar municipalities	399
T3	Out-coalition	Plain info benchmarked by opposition-led municipalities	376
T4	Same-coalition	Plain info benchmarked by same-coalition municipalities	380

- Comparisons matched on population, area, and distance
 - ▶ Consistent with respondent preferences

Information treatment text

[Plain Info]

“Municipal presidents control police finances and, therefore, have partial influence over local crime.”

“In [*municipality*] there were [*X*] robberies per 100,000 people in 2025.”

[Comparative Info]

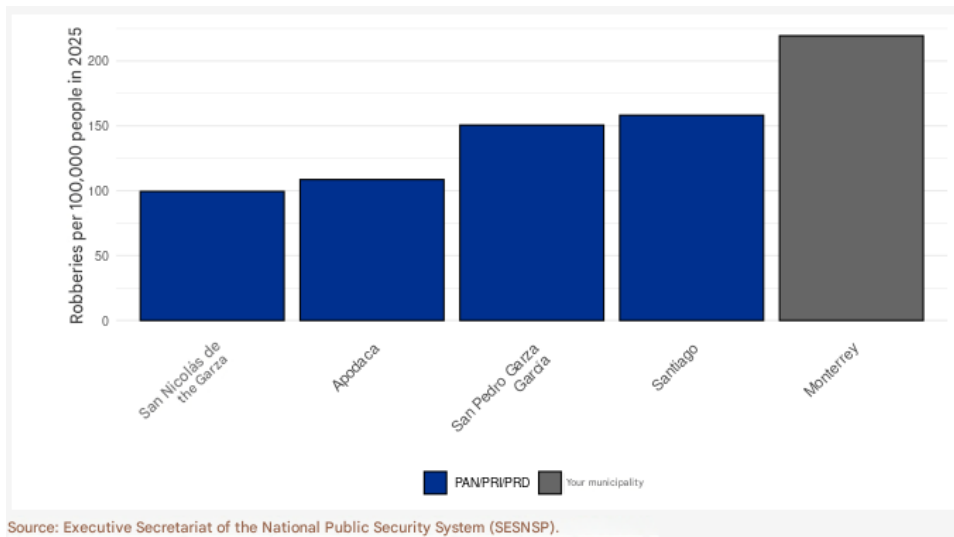
“The following graph shows the robbery rate per 100,000 people in 2025 for [*municipality*] and a sample of similar municipalities...”

T2 (no qualifier)

T3 governed by [*opposite coalition*]

T4 governed by [*same coalition*]

Comparison treatment bar graph example



Perception gaps capture content of information

- Treatment effects should differ based on respondent priors
- Define **perception gap** for respondent i :

$$G_i = Z_i^{\text{actual}} - Z_i^{\text{prior}}$$

- Z_i^{actual} : actual crime data for respondent i 's municipality
- Z_i^{prior} : respondent i 's pre-treatment beliefs about their municipality's crime
- $G_i > 0$: news is worse than expected

Two gaps: crime rate gap (CG_i), and relative ranking gap (RG_i)

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50		
Rank among 4 peers (1 = safest)			

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50	100	
Rank among 4 peers (1 = safest)			

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50	100	$CG_i = +50$
Rank among 4 peers (1 = safest)			

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50	100	$CG_i = +50$
Rank among 4 peers (1 = safest)	5th		

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50	100	$CG_i = +50$
Rank among 4 peers (1 = safest)	5th	2nd	

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

Perception gap examples

	Prior	Actual	Gap
Robberies per 100k	50	100	$CG_i = +50$
Rank among 4 peers (1 = safest)	5th	2nd	$RG_i = -3$

Positive: the news is *worse* than this respondent expected.

Negative: the news is *better* than this respondent expected.

For vote intention outcome y_i :

$$y_i = \alpha + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=1}^4 \delta_k T_{ik} + \sum_{k=1}^4 \beta_k (CG_i \times T_{ik}) + \sum_{k=1}^4 \gamma_k (RG_i \times T_{ik}) + \mathbf{X}_i' \lambda + \varepsilon_i$$

- $T_{ik} = 1$ indicates respondent i is assigned to arm k (1 = plain info, 2 = non-partisan, 3 = opposite-coalition, 4 = same-coalition; control is baseline)
- CG_i : crime rate perception gap for respondent i (signed-logged)
- RG_i : peer ranking perception gap for respondent i
- $\mathbf{X}_i$: pre-treatment controls

For vote intention outcome y_i :

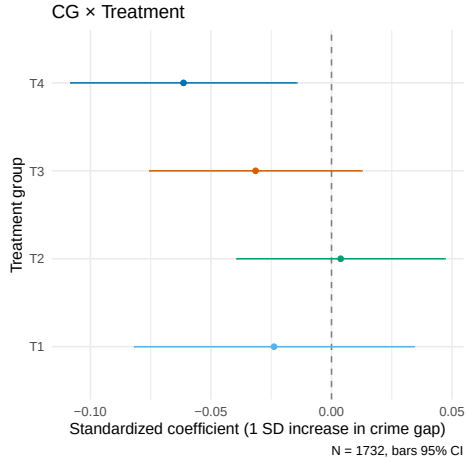
$$y_i = \alpha_0 + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=1}^4 \delta_k T_{ik} + \underbrace{\sum_{k=1}^4 \beta_k T_{ik} \times CG_i}_{\text{Treatment effect from crime rate}} + \underbrace{\sum_{k=1}^4 \gamma_k T_{ik} \times RG_i}_{\text{Treatment effect from ranking}} + \mathbf{X}_i' \lambda + \varepsilon_i$$

For vote intention outcome y_i :

$$y_i = \alpha_0 + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=1}^4 \delta_k T_{ik} + \underbrace{\sum_{k=1}^4 \beta_k T_{ik} \times CG_i}_{\text{Treatment effect from crime rate}} + \underbrace{\sum_{k=1}^4 \gamma_k T_{ik} \times RG_i}_{\text{Treatment effect from ranking}} + \mathbf{X}_i' \lambda + \varepsilon_i$$

Expectation: $\beta_k < 0$ meaning bad news lowers votes for the incumbent

Vote intention for incumbent party



Post-treatment vote intention for the incumbent party (binary) on pre-treatment belief gaps $\times$ arm, vs. control baseline. CG: level gap (actual rate - estimate).

Voting Results Summary

- Only same-coalition comparisons significantly change vote intention
 - ▶ Underestimating crime by SD → **5.3 pp decreased** vote intention for incumbent
 - ▶ Median respondent was off by ≈ 445 per 100,000, roughly one SD from accurate
- Consistent with voters being focused on competence
 - ▶ Theory also predicts opposition benchmark prior-dependent
 - ▶ Test this by looking at RG_i

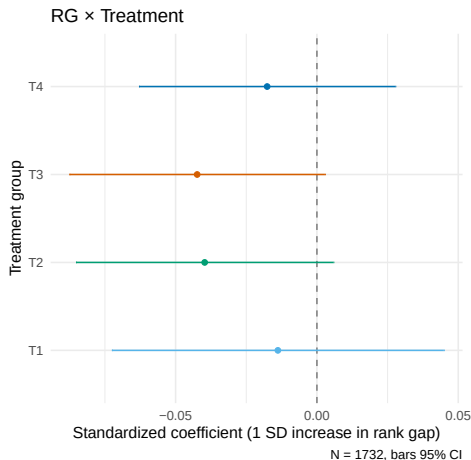
Treatment Effect Estimating Equation

For outcomes y_i :

$$y_i = \alpha_0 + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=1}^4 \delta_k T_{ik} + \underbrace{\sum_{k=1}^4 \beta_k T_{ik} \times CG_i}_{\text{Treatment effect from crime rate}} + \underbrace{\sum_{k=1}^4 \gamma_k T_{ik} \times RG_i}_{\text{Treatment effect from ranking}} + \mathbf{X}_i' \lambda + \varepsilon_i$$

Expectation: $\gamma_k < 0$ meaning bad news lowers votes for the incumbent

Vote intention for incumbent party



Post-treatment vote intention for the incumbent party (binary) on pre-treatment belief gaps × arm, vs. control baseline.

Voting Results Summary (cont.)

- No significant effect from relative perception gaps
- Theory predicts voters interested in competence would be moved
 - ▶ Filtering to respondents who correctly identified home party, effect is larger and statistically significant (no other arm is) Subgroup
- Voting findings are robust to alternative specifications Alt Specs

Additional findings

- Only same-coalition benchmark significantly changes beliefs about incumbent crime handling relative to other coalitions
- Mediation analysis suggests partial mediation of vote outcome through beliefs (pre-registered as exploratory, underpowered)

What do these results mean?

- Votie intentions results imply voters focus on **competence**
 - ▶ Other interpretations of benchmarks not supported
- Benchmarks are design choices, not minor presentation details
 - ▶ Non-partisan comparisons are the field's default, but filter the least noise
 - ▶ Same-party benchmarks uncommon, but are well suited in many cases

Thank you!

Adam D. Roberts

<https://adamdnroberts.github.io>



QR code to full paper

Part I

Appendix

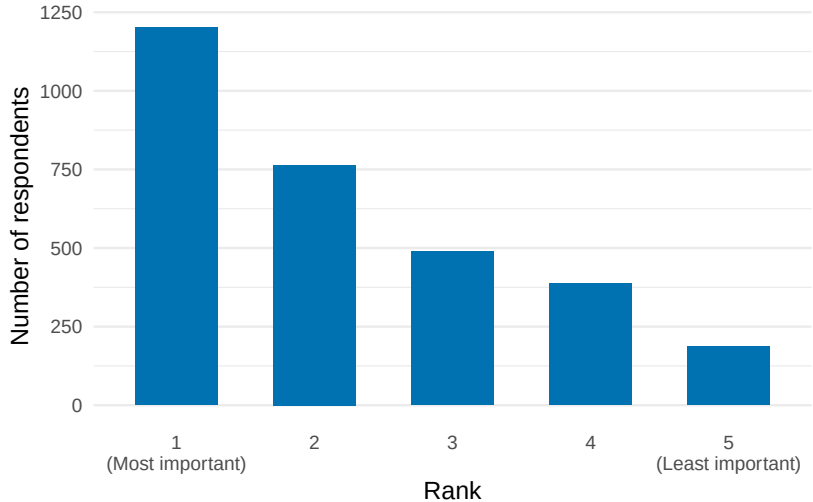
Part content

Why crime information?

- Municipal forces comprise majority of law enforcement (Marshall 2024)
- Municipal presidents control police funding, hiring, and structure
- Crime is a top voter concern for Mexicans
- Imprecise priors about local crime → information should update beliefs

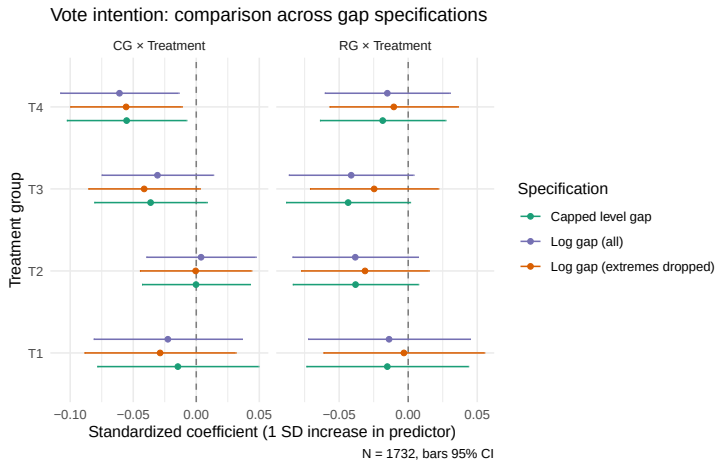
Data: robbery rate per 100,000 people in 2025, from Public Security Secretary (SESNSP)

Crime is a top voter concern for Mexicans



N = 3028

Alternative crime-gap specifications

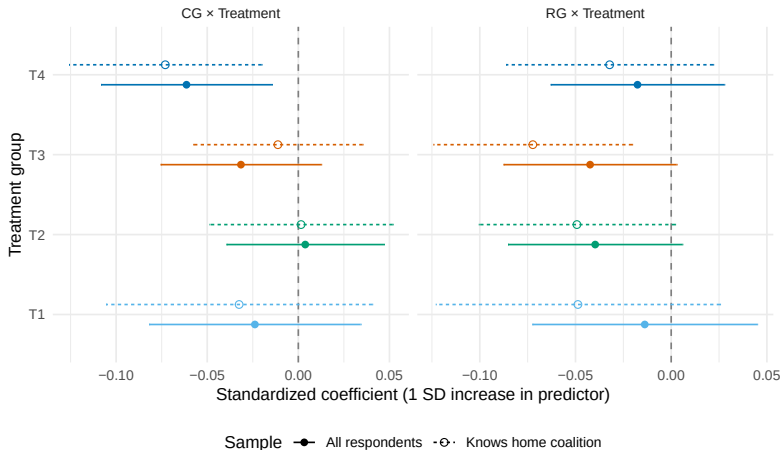


Treatment interaction coefficients for post-treatment incumbent vote intention under three treatments of the crime gap: capped level gap, log gap, and log gap with extreme estimates dropped. $CG \times T4$ is the only crime-gap interaction distinguishable from zero under every specification (-0.061, -0.055, -0.056). $N = 1,732$; $N = 1,604$ for extremes-dropped.

Capped CG

Back

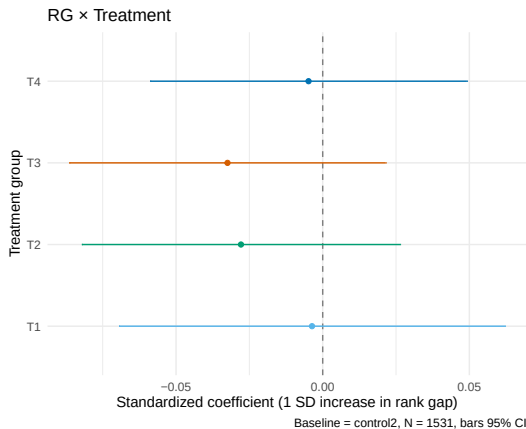
Filtering to knowledge of home municipality



N = 1732 (all) / 1376 (knows coalition), bars 95% CI

Interaction coefficients in the full estimation sample and among respondents who correctly named their home municipality's governing coalition. $CG \times T4$ goes from -0.061 ($p = 0.010$) to -0.073 ($p = 0.007$); $RG \times T3$ from $p = 0.066$ to -0.072 ($p = 0.006$). Scaled by full-sample SDs. $N = 1,732$ and $N = 1,376$.

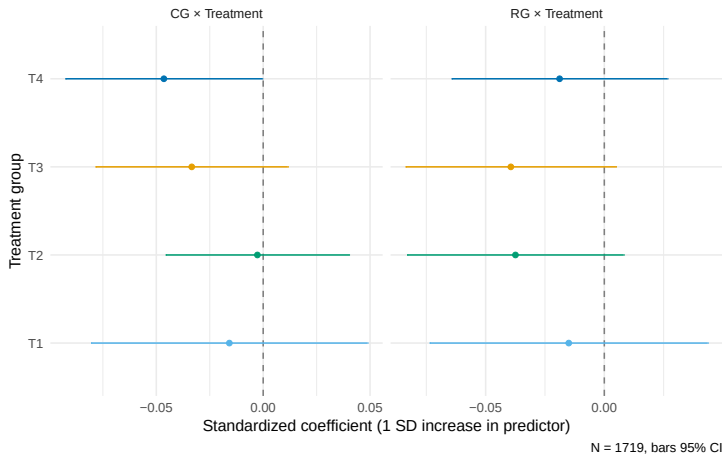
Vote intention with control2 as the baseline



RG × arm interactions benchmarked against control2 (placebo text plus comparison chart) rather than control, so the estimates net out the presence of a comparison. For control2, the rank gap is built from the rainfall figures those respondents were shown.

Back

Vote intention updates with capped CG specification



Same model as the main vote specification, but the crime gap enters as the capped level gap rather than the log gap.

Back